

F78PB Probability and Statistics B

Hypothesis Testing

Hypothesis Testing

Until now, we have developed methods for estimating an unknown parameter θ using:

- Point estimation
- Confidence interval estimation

Now we consider a different question: starting with a **hypothesis**.

Example

Imagine a company claims that a particular model of its mobile phone lasts **150 hours** before needing to be charged.

One way to check this assertion is to take a sample of phones and calculate the mean time between charges. Testing every phone is impossible, so we rely on a sample mean.

A sample of $n = 40$ phones produced:

$$\bar{x} = 147.4 \text{ hours}, \quad s = 12 \text{ hours}.$$

Question: Does this mean the manufacturer's claim is false? Clearly $147.4 < 150$, so it seems the claim might be an overestimate—but is this difference statistically significant?

Why Hypothesis Testing?

Our sample result (147.4 hours) comes from just one sample. Another sample might give a very different result (e.g., 152.3 hours), suggesting the phones perform better than claimed.

The method of **hypothesis testing** begins by making an assertion about the population— e.g. that the mean equals a stated value.

In this case, we hypothesize that the population mean μ for the mobile phones is 150 hours.

Setting Up the Hypotheses

Method of Solution:

- Assert that the population mean is what the company claims. This is the **Null Hypothesis**, H_0.
- Perform an analysis to decide whether to reject or not reject this assertion.

The hypotheses are:

$$H_0 : \mu = 150 \qquad \text{versus} \qquad H_1 : \mu \neq 150,$$

where H_1 is the Alternative Hypothesis.

Using theory developed before, assuming H_0 to be true implies that

$$\frac{\bar{X} - 150}{\frac{S}{\sqrt{n}}} \dot{\sim} \mathrm{N}(0, 1)$$

For our particular sample we can calculate a value for

$$\frac{\bar{x} - 150}{\frac{s}{\sqrt{n}}} = \frac{147.7 - 150}{12/\sqrt{40}} = -1.370$$

Now recall that for a $\mathrm{N}(0, 1)$ distribution, 95% of its values, say, are between -1.96 and 1.96.

So the observed value -1.370 (based on the sample) looks like a typical value from this distribution.

Therefore we have insufficient evidence to reject the null hypothesis that the mean time between charging for this type of phone is 150 hours.

Hypotheses

A **statistical hypothesis** is an assertion about the distribution of one or more random variables.

The basic hypothesis to be tested is the Null Hypothesis H_0. It expresses knowledge or belief about an unknown parameter before observing data.

To test H_0, we also need an Alternative Hypothesis H_1.

Types of hypotheses:

- Simple hypothesis: Completely specifies the distribution (e.g., $\mu = 150$).
- Composite hypothesis: Does not fully specify the distribution (e.g., $\mu > 150$).

Test Statistics and Decision Rule

A **statistical test** is a rule that divides the sample space into two regions:

- Acceptance region: Values consistent with H_0.
- Rejection (critical) region: Values inconsistent with H_0.

The test is based on a **test statistic**, which summarizes the evidence against H_0.

Decision:

- If the test statistic falls in the rejection region $\rightarrow$ **Reject** H_0.
- Otherwise $\rightarrow$ **Do not reject** H_0.

The decision regarding the test is based on the value of the test statistic, which is a suitable function of the data and the parameter(s) being tested (and its distribution is known). The range of values of a test statistic is also divided into two regions:

-an acceptance region where the values of the statistic are consistent with H_0

-and its complement the rejection region where the values of the statistic are inconsistent with H_0.

Errors Associated with Statistical Tests

The probability of a Type I Error is the probability to **reject** H_0 when it is **true**. It is denoted by α and is also referred to as the significance level of the test.

$$\mathbb{P}(\text{type I error}) = \alpha$$

The probability of a Type II Error is the probability to **fail to reject** H_0 when it is **false**. This is denoted by β.

Test Statistic and Decision Regions

The decision regarding the test is based on the value of the **test statistic**, a function of the data and the parameter(s) being tested, whose distribution is known.

The range of values of the test statistic is divided into two regions:

- Acceptance region: Values consistent with H_0.
- Rejection (critical) region: Values inconsistent with H_0.

Errors Associated with Statistical Tests

Type I Error: Reject H_0 when it is **true**.

$$\mathbb{P}(\text{Type I error}) = \alpha,$$

where α is the **significance level** of the test.

Type II Error: Fail to reject H_0 when it is **false**.

$$\mathbb{P}(\text{Type II error}) = \beta.$$

Note: Reducing α usually increases β, and vice versa.

Also define the **power** of a test as:

$$\mathbb{P}(\textbf{reject } H_0 \text{ when it is } \textbf{false}) = 1 - \beta.$$

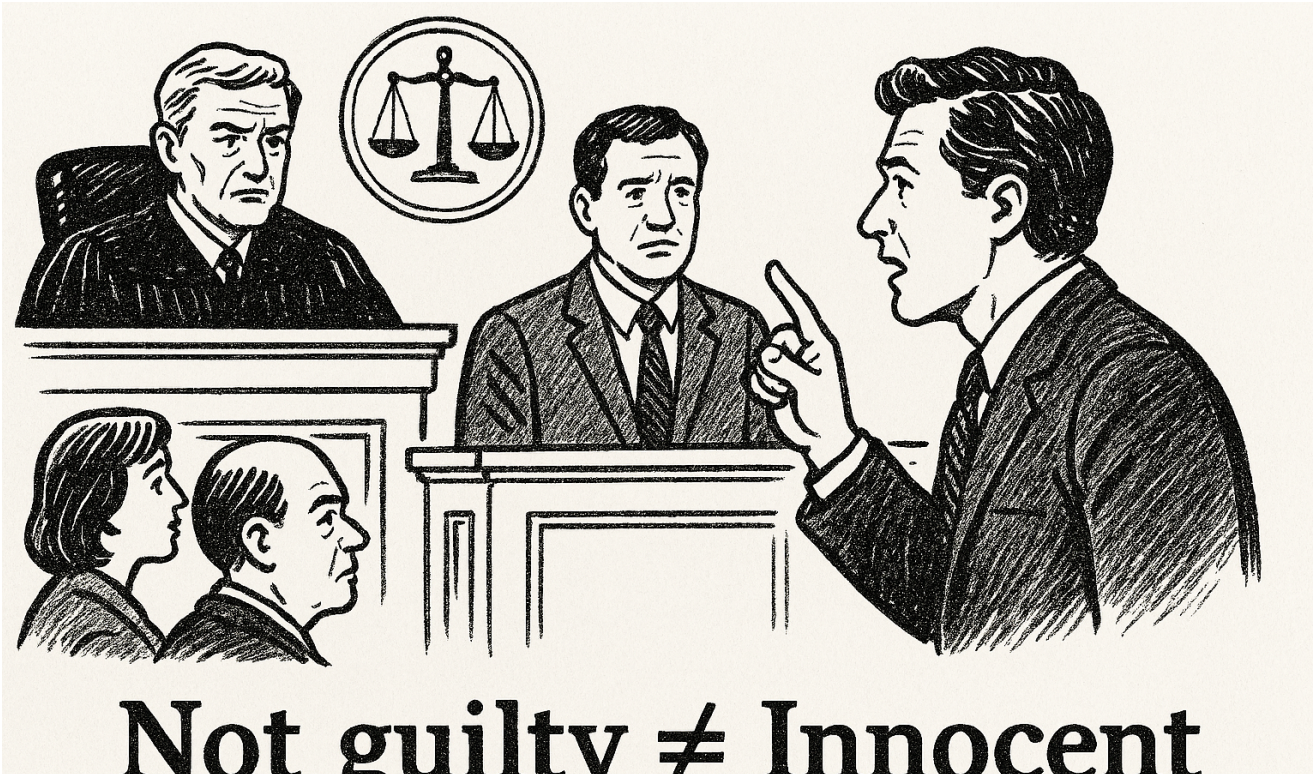
Ideally we wish to minimise both α and β simultaneously - not achieved in practice (for fixed sample size n). More on errors and power of tests in later statistics courses.

Failing to Reject vs Accepting H_0

- **Failing to reject** H_0: insufficient evidence against H_0.
- **Accepting** H_0: concluding H_0 is true.
- These are **not equivalent** in hypothesis testing.

Analogy

Failing to reject H_0 is like a jury saying "not guilty"—it does not mean "innocent", only that there was insufficient evidence to convict.



Example: Why Non-Rejection $\neq$ Acceptance

Consider the test:

$$H_0 : X \sim \mathrm{N}(0,1) \quad \text{vs} \quad H_1 : X \text{ is not distributed according to } \mathrm{N}(0,1).$$

If H_0 is true, $\mathbb{P}(-3 \leq X \leq 3) \approx 0.997$, so reject H_0 if $X \leq -3$ or $X \geq 3$.

Now suppose $X \sim \mathrm{Ber}(0.5)$. Then X never falls outside $[-3, 3]$, so we do **not reject** H_0. Does this imply $X \sim \mathrm{N}(0,1)$?

Clearly not.

Example

A random sample of size $n = 9$ is taken from a $\mathrm{N}(\mu, 900)$ population to test the hypothesis

H_0: $\mu = 100$ v H_1: $\mu \neq 100$
at the 5% level of significance ($\alpha = 0.05$).

(a) Find the critical (rejection) region of this test in terms of $\bar{X}$.

(b) Calculate the probability of a Type II error of this test when $\mu = 105$.

Example

In a sequence of $n = 20$ independent trials we want to test the hypothesis
$H_0 : p = 0.9$ v $H_1 : p = 0.6$
where p is the probability of 'success'. The test is based on the number of successes, X in n trials and we reject H_0 when $X \leq 14$.

Find the probabilities of type I and type II errors.

Probability Value (p-value) of a Test

In the mobile phone example, the decision (conclusion) would be that we do not reject H_0 at the 5% significance level.

This does not describe in full detail the strength of the evidence in favour (or against) H_0.

A more informative approach is based on the following:

The p-value corresponding to an observed value of a test statistic, is the **probability to observe a test statistic as or more extreme as the value actually observed** (under the assumption that H_0 is true).

Also referred to as the tail probability or the observed significance level of the test, it gives the **lowest level of significance at which H_0 could have been rejected**.

Hypothesis Testing in Practice

Once appropriate data are given, we can perform a statistical test for some population parameter following the steps below:

1. Formulate H_0 and H_1 and specify the significance level α.
2. Select an appropriate test statistic.
3. Calculate the value of the test statistic under H_0 using the data.
4. Either:
   1. Determine a critical region of size α according to the chosen test statistic Or
   2. Calculate the p-value that corresponds to the value of the test statistic.
   - The critical region depends on the sampling distribution of the test statistic, the significance level α and the nature of the alternative hypothesis H_1 (one or two-tailed test).
   - The p-value is the probability to observe a value of the test statistic as or more extreme as the value obtained from the sample data.
5. Reject the null hypothesis if
   Either the value of the test statistic falls into the critical region Or the p-value is less than or equal to the significance level α.
   Otherwise do not reject H_0 (or reserve judgement).

Hypothesis Testing For Means

Single sample situation

Test H_0: $\mu = \mu_0$
versus H_1: $\mu \neq \mu_0$ (or $\mu > \mu_0$, etc.)
at a specified significance level α.
With normal samples or large sample size n, use the 'z test' statistic

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim \mathrm{N}(0, 1)$$

If σ is unknown and n is large use the sample standard deviation, s instead.
Otherwise, for small, normal samples we perform a t-test, based on the test statistic

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \quad \sim t_{n-1}$$

Example

According to its specifications, the mean breaking strength of a certain kind of ribbon is 185 pounds. Five pieces are randomly selected and have breaking strengths (in pounds)

171.6 191.8 178.3 184.9 189.1

a. Test the hypothesis that the mean is 185 pounds against the alternative that it is less than 185 pounds, at the 5% level of significance.
b. Calculate the p-value for this test and give its interpretation.
c. Now consider a test of H_0: $\mu = 185$ v H_1: $\mu \neq 185$ and repeat the exercise.